

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
PART TEST – III
PAPER –1
TEST DATE: 24-12-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 03, 18 – 20, 35 – 37): This section contains **NINE (9)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (04 – 07, 21 – 24, 38 – 41): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (08 – 11, 25 – 28, 42 – 45): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (12 – 17, 29 – 34, 46 – 51): This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

SECTION – A

(One or More than one correct type)

This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. A large steel cylinder has inner radius 'a' as shown in the figure-(ii). There is a cork of radius ' $a + \Delta a$ ' ($\Delta a \ll a$) and length 'L' shown in the figure (i). Now, cork is to be fitted to close the open mouth of cylinder. Consider poisson's ratio of cork is almost zero and its bulk modulus is 'B'. Choose the correct option(s).

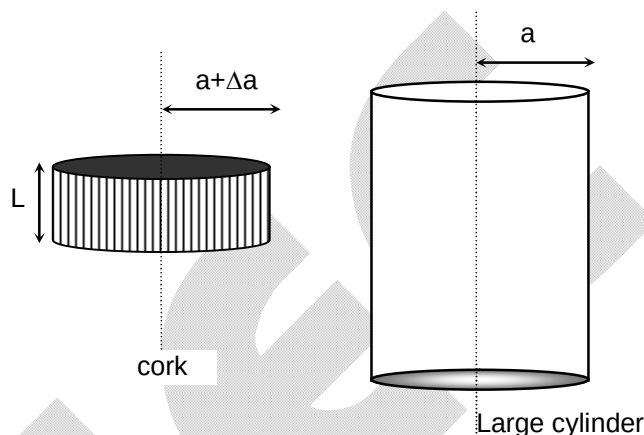
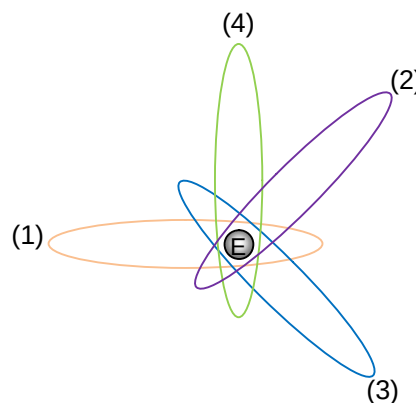


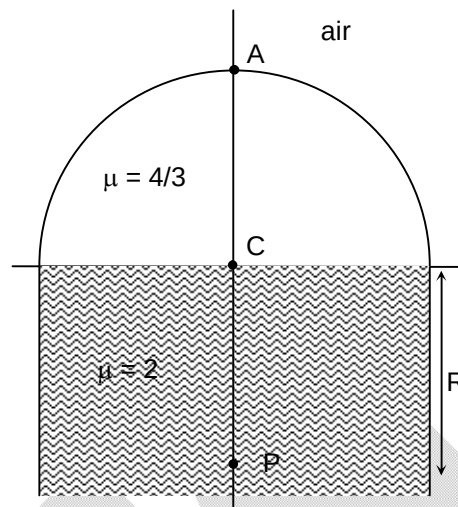
Figure (i)

Figure (ii)

- (A) Steel is less elastic than cork
 (B) An uniform pressure on curved cylindrical surface of cork is developed which is equal to $\frac{2B\Delta a}{a}$
 (C) If ' μ ' is the coefficient of friction between cork and container then the needed force to pull it out of the container is $2\pi\mu BL\Delta a$.
 (D) When stress is applied on a wire is increased beyond the yield point and is removed after some time then the wire can not return to its original position.
2. There are four identical artificial satellites orbiting around the Earth in four elliptical orbits having semi-major axis of equal length but different eccentricity, assuming earth is at rest. Pick the correct statement(s) regarding motion of satellite.
- (A) The magnitude of maximum and minimum velocity of satellite in each orbit is equal
 (B) Total energy of satellite in each orbit is same and constant
 (C) Angular momentum of satellite in each orbit is same and constant
 (D) Time period of satellite in each orbit is same



3. A luminous point object P is kept in a medium of refractive index $\mu = 2$ at a distance 'R' from the centre of hemisphere of refractive index $\mu = \frac{4}{3}$ and radius of curvature R as shown in the figure. Assuming paraxial approximation. Choose the correct statement(s).

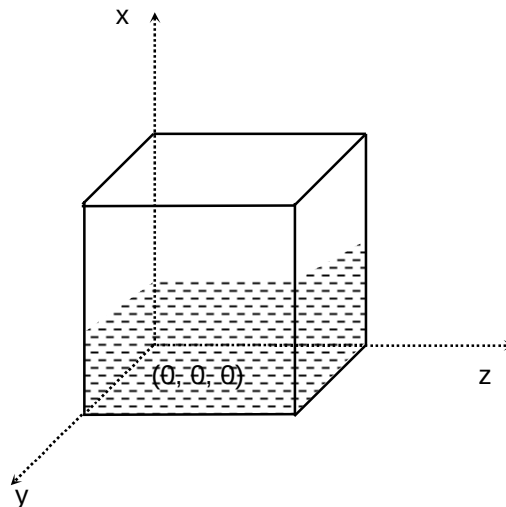


- (A) Magnitude of the change in radius of curvature of the wave front just after it enter at point C is $\frac{R}{3}$.
- (B) Magnitude of the change in radius of curvature of the wave front just after it enter at point C is $\frac{2R}{3}$.
- (C) Radius of curvature of the wave front at point A just outside the hemisphere is $\frac{15}{17}R$.
- (D) Radius of curvature of the wave front at point A just outside the hemisphere is $\frac{15}{7}R$.

SECTION – A
(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

4. A cubical container is filled with liquid of density 'ρ' upto a height H. Container is moving with constant acceleration $a_1\hat{i} + a_2\hat{k}$. Acceleration due to gravity is along negative x-direction. Container is open at the top and assume liquid is not spill out. Consider the pressure P_0 at the origin. The pressure at co-ordinate (x, y, z) inside the liquid is



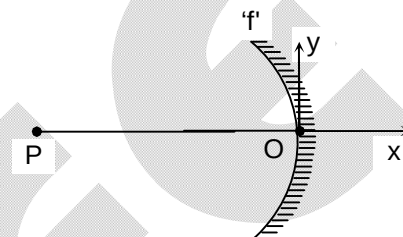
- (A) $P_0 - \rho a_1 x - \rho g y - \rho a_2 z$
- (B) $P_0 - \rho(g + a_1)z - \rho a_2 x$
- (C) $P_0 - \rho(g + a_1)x - \rho a_2 z$
- (D) $P_0 + \rho(g + a_1)x + \rho a_2 z$

5. The radiation power of a star is given by $P = P_0 \left(\frac{m}{M_s} \right)^{9/2}$, where P_0 is radiation power of sun, M_s is mass of the sun and m is mass of star at time t . Consider initial mass of star is m_0 . The mass of star at time ' t ' is best represented by

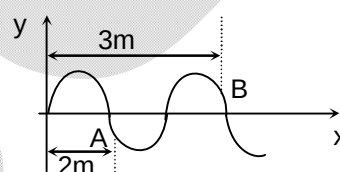
(A) $\frac{1}{m^{7/2}} = \frac{1}{m_0^{9/2}} + \frac{7P_0 t}{2(M_s)^{9/2} c^2}$ (B) $\frac{1}{m^{7/2}} = \frac{1}{m_0^{7/2}} + \frac{7P_0 t}{2(M_s)^{9/2} c^2}$
 (C) $\frac{1}{m^{9/2}} = \frac{1}{m_0^{7/2}} + \frac{7P_0 t}{2(M_s)^{7/2} c^2}$ (D) $\frac{1}{m^{9/2}} = \frac{1}{m_0^{9/2}} + \frac{7P_0 t}{2(M_s)^{7/2} c^2}$

6. A concave mirror has a focal length $f = 15$ cm. A point object is kept at its centre of curvature. Now if mirror is rotated about z -axis through a very small angle $\theta = 2^\circ$. Consider paraxial rays only. The approximate distance between object and its image is

(A) $\frac{\pi}{150}$ m (B) $\frac{\pi}{75}$ m
 (C) $\frac{\pi}{300}$ m (D) $\frac{\pi}{100}$ m



7. A transverse wave is propagating in stretched string along positive x -direction. The density of medium is 450 kg/m^3 . The oscillation energy density of particle in the medium is $0.25 \pi^2 \text{ J/m}^3$ and maximum shear strain produced in the medium is $3\pi \times 10^{-6}$. If the phase difference between particle A and B is 72° as shown in the figure then the equation of wave is best represented by

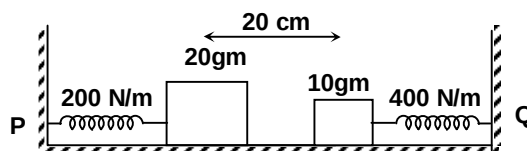


(A) $\left(\frac{15}{2} \times 10^{-6} \right) \sin \left(\frac{20000}{9} \pi t - \frac{2\pi}{5} x \right)$ (B) $\left(15 \times 10^{-6} \right) \sin \left(\frac{40000}{9} \pi t - \frac{\pi}{5} x \right)$
 (C) $\left(15 \times 10^{-6} \right) \sin \left(\frac{20000}{9} \pi t - \frac{2\pi}{5} x \right)$ (D) $\left(\frac{15}{2} \times 10^{-6} \right) \sin \left(\frac{40000}{9} \pi t - \frac{2\pi}{5} x \right)$

SECTION – A (Matching List Type)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

8. Two small block of mass 10 gm and 20 gm are connected with two ideal spring of force constant 400 N/m and 200 N/m respectively. Springs are fixed at Q and P. Horizontal surface is smooth. The natural length of each spring is 20 cm and the distance between point P and Q is 40 cm. Initially the blocks are at 20 cm and compression in both spring is identical. Now, blocks are released, they collides perfectly inelastically during their motion.



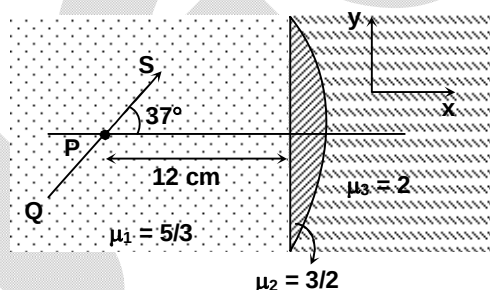
Match each entry in List-I with an appropriate value from List-II and choose correct option.

List -I		List -II	
(P)	Velocity of block of mass 20 gm just before collision in m/s	(1)	$5\sqrt{2}$
(Q)	Displacement of block of mass 10 gm just before collision in meter	(2)	Zero
(R)	Momentum of combined blocks just after collision in kg-m/s	(3)	0.75
(S)	Maximum speed of combined block during oscillation in m/s	(4)	0.15
		(5)	$5\sqrt{3}$

The correct option is:

- (A) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (3) (R) $\rightarrow$ (4) (S) $\rightarrow$ (1)
 (B) (P) $\rightarrow$ (5) (Q) $\rightarrow$ (4) (R) $\rightarrow$ (2) (S) $\rightarrow$ (1)
 (C) (P) $\rightarrow$ (4) (Q) $\rightarrow$ (3) (R) $\rightarrow$ (2) (S) $\rightarrow$ (1)
 (D) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (4) (R) $\rightarrow$ (1) (S) $\rightarrow$ (5)

9. A plano-convex lens of refractive index $\mu_2 = \frac{3}{2}$ is kept in two different medium of refractive index $\mu_1 = \frac{5}{3}$ and $\mu_3 = 2$ as shown in the figure. The radius of curvature of curved surface of planoconvex lens is 12 cm. An object is performing simple harmonic motion with amplitude 5 mm along a line QPS with mean position at P. The maximum velocity of particle during the motion is 1 m/s.



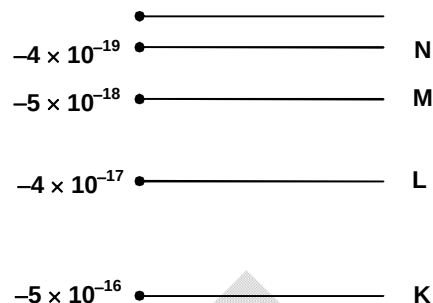
Match each entry in List-I with an appropriate value from List-II and choose the correct option.

List -I		List -II	
(P)	Amplitude of oscillation of image along x-axis in mm	(1)	$\frac{30}{13}$
(Q)	Amplitude of oscillation of image along y-axis in mm	(2)	$\frac{6}{13}$
(R)	Maximum velocity of image along x-axis in m/s	(3)	$\frac{120}{169}$
(S)	Maximum velocity of image along y-axis in m/s	(4)	$\frac{96}{169}$
		(5)	$\frac{480}{169}$

The correct option is:

- (A) (P) $\rightarrow$ (5) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (3) (S) $\rightarrow$ (2)
 (B) (P) $\rightarrow$ (4) (Q) $\rightarrow$ (2) (R) $\rightarrow$ (3) (S) $\rightarrow$ (1)
 (C) (P) $\rightarrow$ (5) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (4) (S) $\rightarrow$ (2)
 (D) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (2) (R) $\rightarrow$ (1) (S) $\rightarrow$ (4)

10. Consider a hypothetical atom, the energy (in J) levels of atoms is given in the figure. The atom is used as a target metal to produce X-rays. High energetic electrons are incident on target metal. Let us consider a vacancy is created in K-level, this vacancy in K-level is filled by an electron from L-level and the energy released in this transition can either appear as X-rays or may all be used to knock out an electron from M-level of the atom.

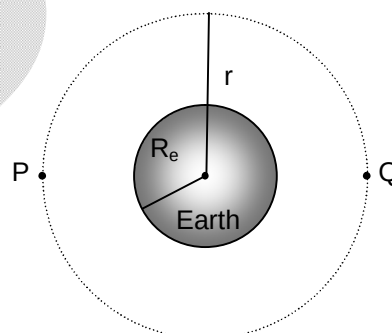


Match each entry in List-I with an appropriate value from List-II and choose the correct option.

List -I		List -II	
(P)	Required minimum potential difference (in volt) of incident electrons for emission of electron from K-level	(1)	56.7×10^{-10}
(Q)	Wavelength (in meter) of L_{α} type X-rays	(2)	455×10^{-18}
(R)	Frequency of K_{α} type X-rays	(3)	6.3×10^{12}
(S)	Kinetic energy (in J) of emitted electron from M-level	(4)	6.95×10^{17}
		(5)	3.125×10^3

The correct option is:

- (A) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (2) (R) $\rightarrow$ (1) (S) $\rightarrow$ (4)
 (B) (P) $\rightarrow$ (5) (Q) $\rightarrow$ (4) (R) $\rightarrow$ (3) (S) $\rightarrow$ (2)
 (C) (P) $\rightarrow$ (4) (Q) $\rightarrow$ (3) (R) $\rightarrow$ (2) (S) $\rightarrow$ (1)
 (D) (P) $\rightarrow$ (5) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (4) (S) $\rightarrow$ (2)
11. Two satellites 'P' and 'Q' are moving in a circular orbit about the Earth. The masses of both satellites are same and equal to 500 kg. The radius of circular orbit is 9600 km. Satellite 'P' is moving in clockwise direction and Q anticlockwise. Both satellites undergo a collision which is perfectly inelastic. Assuming earth is at rest. Given mass of earth = 6×10^{24} kg, radius of earth = 6400 km and Gravitational constant $G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$.



Match each entry in List-I with an appropriate value from List-II and choose the correct option.

List -I		List -II	
(P)	Time period of satellite before collision in sec	(1)	30.99×10^{12}
(Q)	Magnitude of total energy of system before collision in joule	(2)	15.68×10^6
(R)	Angular momentum of each satellite before collision in $\text{kg-m}^2/\text{sec}$	(3)	20.84×10^9
(S)	Magnitude of total energy of system just after collision.	(4)	9.34×10^3
		(5)	41.68×10^9

The correct option is:

- (A) (P) $\rightarrow$ (1) (Q) $\rightarrow$ (3) (R) $\rightarrow$ (5) (S) $\rightarrow$ (4)
 (B) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (3) (R) $\rightarrow$ (4) (S) $\rightarrow$ (1)
 (C) (P) $\rightarrow$ (4) (Q) $\rightarrow$ (3) (R) $\rightarrow$ (1) (S) $\rightarrow$ (5)
 (D) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (5) (S) $\rightarrow$ (4)

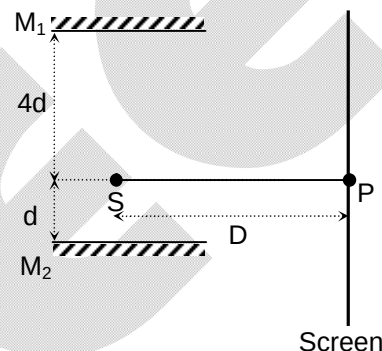
SECTION – B

(Numerical Answer Type)

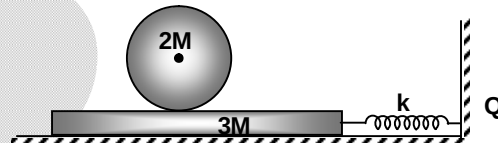
This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

12. Consider a cylinder of height H and radius R . The side wall of container has very small holes which are uniformly distributed to its entire surface area. The number of holes are very large and occupying a fraction $k = 0.003$ of the entire area of the wall. Water is being poured to container at a constant rate Q . To maintain the water level at height $H/2$, the value of Q is $\frac{\pi}{\lambda} R \sqrt{g} H^{3/2}$. Find the value of λ . (Assume torricelli's law is valid)

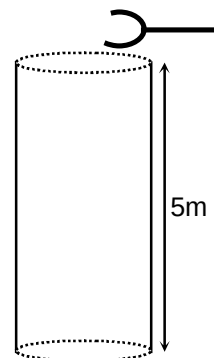
13. M_1 and M_2 are two fixed plane mirrors. A luminous point object 'S' is kept at distances d and $4d$ from mirror M_2 and M_1 respectively as shown in the figure. An interference pattern is obtained by the first image of both the mirrors on a screen kept at a distance ' D ' from the source 'S'. It is observed that 2nd maxima is formed at point 'P' on the screen. It is given that $D = 3$ m and $d = 1$ mm. If monochromatic light of wave length $\lambda = k$ nm is used then find the value of k .



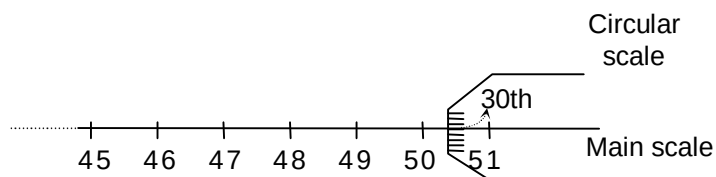
14. A large plank of mass ' $3M$ ' is kept on a frictionless horizontal surface, one end of plank is connected with a spring of force constant $k = 110$ N/m as shown in the figure. A cylinder of mass ' $2M$ ' and radius ' R ' is placed on the upper surface of plank. Friction force is sufficient to prevent slipping between cylinder end plank. Now plank is slightly displaced from its equilibrium position, then its angular frequency of oscillation is given by $\sqrt{N}$ rad/s. Find the value of N . (Given $M = 1$ kg)



15. A cylindrical open organ pipe of length 5 m is used to produce a stationary wave with the help of a tuning fork as shown in the figure. Standing wave vibrates in its fifth harmonic. The amplitude of pressure oscillation is one percent of mean atmospheric pressure. Velocity of sound in the medium is 330 m/s. Density of air is equal to 1.04 kg/m^3 and atmospheric pressure is 10^5 N/m^2 . End correction is neglected. The amplitude of particle density oscillation is $\frac{10}{N} \text{ kg/m}^3$. Find the value of N .



16. The diameter of a thin wire is measured by using a screw gauge having 100 divisions on circular scale. When spindle of screw gauge is touched to anvil then zero of circular scale lies 6 division above the reference line of main scale.



- scale. Now, when wire is kept between spindle and anvil then 30th division of circular scale exactly coincides with the reference line of main scale as shown in the figure. The distance between two successive divisions of main scale is 0.1 mm. Find the measured diameter of wire in micrometer.

17. A light beam of intensity $5.6 \times 10^{-4} \text{ W/m}^2$ is used to emit the photoelectron from a metallic surface of work function 2.32 eV. Light beam has four wavelengths 4532 Å, 5112 Å, 5628 Å and 6123 Å. Intensity of light beam is equally distributed amongst the four wavelengths. Light beam falls normally on an area of 8.0 cm^2 . Efficiency of emission of electron from metallic surface is 40%. The number of emitted photoelectrons from metallic surface in 3 sec is 6.54×10^9 . Find the value of p . ($hc = 12400 \text{ eV-Å}$)

Chemistry

PART – II

SECTION – A

(One or More than one correct type)

This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

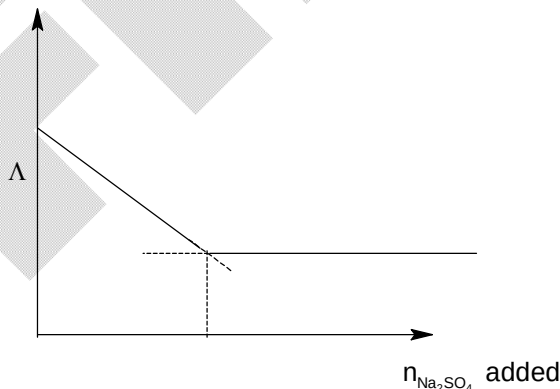
18. Bauxite containing high percentage of silica can be purified by Serpeck's process. In this process finally powdered bauxite is mixed with coke and the mixture is heated to 1800°C in a current of nitrogen. And hot & dil. NaOH is also used. Which of the following is/are correct?
 (A) N_2 remains inert in the complete process.
 (B) Al of the ore ends up as Al_2O_3 in this process, which is reduced electrolytically to obtain Al.
 (C) Al ore ends up as NaAlO_2 , in presence of hot and dil. NaOH.
 (D) Carbon reduces Al_2O_3 to Al, in presence of N_2 .
19. What type of sol will be formed if FeCl_3 is added to a solution prepared by mixing 200 ml, 0.4 M HCl and 800 ml, 0.7 M NH_4OH ?
 (A) Cationic (B) Anionic
 (C) Lyophilic (D) Macromolecular
20. Which of the following is/are correct?
 (A) Cold solution of $\text{Ba}(\text{NO}_2)_2$ on mixing with H_2SO_4 produces BaSO_4 and HNO_2 .
 (B) HNO_3 can not be used for the preparation of H_2S from metal sulphides.
 (C) $\text{P}_4 + \text{NaOH} \xrightarrow{\text{H}_2\text{O}} \text{PH}_3 + \text{NaH}_2\text{PO}_3$
 (D) $\text{Ca}_3(\text{PO}_4)_2$ give H_3PO_4 on reaction with H_2SO_4 .

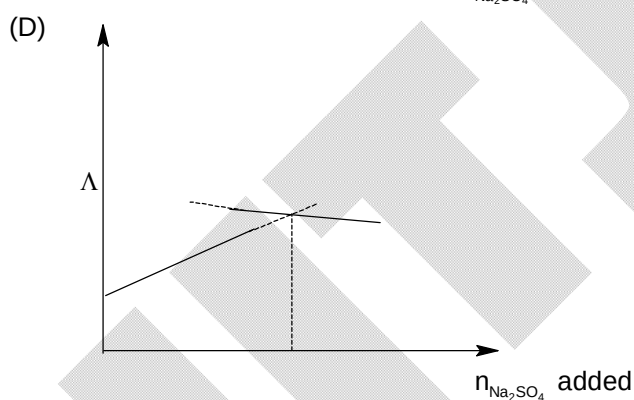
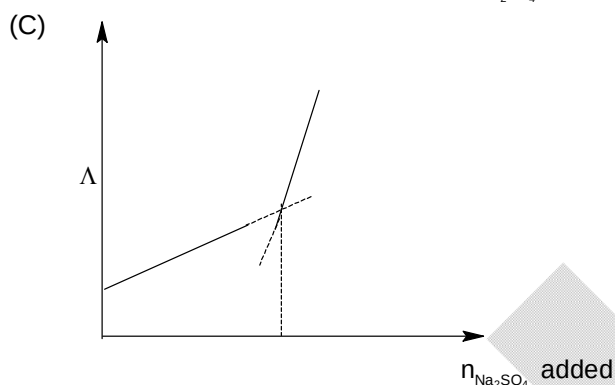
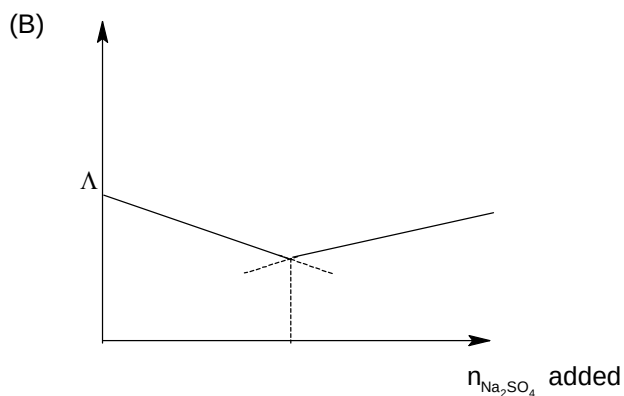
SECTION – A

(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

21. Freshly prepared solution of HClO_2 is colourless but it becomes yellow due to formation of.....?
 (A) Cl_2 (B) HOCl
 (C) HClO_3 (D) ClO_2
22. Which of the following closely represents the variation of conductance of BaCl_2 solution which is gradually contaminated by Na_2SO_4 ?
 (A)





23. Which of the following is correct?

- (A) $\text{F}_2 + \text{KClO}_3 \xrightarrow{\text{H}_2\text{O}} \text{KClO}_2 + \text{HF}$
 (B) $\text{F}_2 + \text{NaHSO}_3 \longrightarrow \text{Na}_2\text{S}_2\text{O}_8 + \text{HF}$
 (C) $\text{Cl}_2 + \text{Na}_2\text{SO}_3 \xrightarrow{\text{H}_2\text{O}} \text{Na}_2\text{S}_2\text{O}_8 + \text{HCl}$
 (D) $\text{Br}_2 + \text{Ca}(\text{OH})_2 \longrightarrow \text{CaBr}_2 + \text{H}_2\text{O}$

24. Which of the following is **incorrect** regarding $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$?

- (A) It can exhibit linkage isomerism
 (B) It can exhibit ionization isomerism
 (C) Its ionization isomers show geometrical isomerism
 (D) One mole of any of its ionization isomers can give 2 mole of white precipitate on reaction with AgNO_3 in aqueous solution

SECTION – A
(Matching List Type)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

25. Match the list given below;

List – I		List – II	
(P)	Sulphaguanidine	(1)	An artificial sweetner
(Q)	Sucralose	(2)	An antihistamine
(R)	Cetyltrimethylammonium bromide	(3)	A detergent
(S)	Terfenadine	(4)	Used in bacillary dysentery
		(5)	Used in pneumonia

- (A) P → 2; Q → 3; R → 4; S → 1
 (B) P → 1; Q → 3; R → 4; S → 2
 (C) P → 4; Q → 1; R → 3; S → 2
 (D) P → 2; Q → 4; R → 1; S → 3

26. Match the following:

List – I		List – II	
(P)	Lead storage battery	(1)	Metal oxide plate is cathode
(Q)	Ni-Cd battery	(2)	Pt metal plate is used as anode and cathode both
(R)	H ₂ – O ₂ fuel cell	(3)	pH decreases at cathode during discharge
(S)	CH ₄ – O ₂ fuel cell	(4)	pH increases at anode during discharge
		(5)	O ₂ is consumed at cathode

- (A) P → 2; Q → 1; R → 2; S → 5
 (B) P → 1; Q → 2; R → 2; S → 5
 (C) P → 1; Q → 1; R → 2; S → 5
 (D) P → 1; Q → 2; R → 1; S → 2

27. Match the following:

List – I		List – II	
(P)	KO ₂	(1)	Paramagnetic
(Q)	K ₃ [Fe(CN) ₆]	(2)	Diamagnetic
(R)	Fe ₃ O ₄	(3)	Ferromagnetic
(S)	MnO	(4)	Ferromagnetic
		(5)	Antiferromagnetic

- (A) P → 2; Q → 1; R → 4; S → 5
 (B) P → 1; Q → 1; R → 4; S → 5
 (C) P → 1; Q → 2; R → 5; S → 4
 (D) P → 2; Q → 1; R → 3; S → 5

28. Match the following:

List – I		List – II	
(P)	Uranium	(1)	Cheralite
(Q)	Cerium	(2)	Monazite
(R)	Thorium	(3)	Thorite
(S)	Lanthanum	(4)	Carnotite
		(5)	Samarskite- (Y)

- (A) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$
 (B) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 5$
 (C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$
 (D) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 2$

SECTION – B

(Numerical Answer Type)

This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

29. AgBr(s) is used in photography to fix the image on photographic plate, in which it forms a complex with hypo. What is the molar mass of the complex ion formed in this process?
[Atomic mass = Ag = 108; Br = 80; S = 32; O = 16]
30. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ is used in fire works. What is the molar mass of the green coloured powder blown in air? (Atomic mass = Cr = 52; O = 16; N = 14; H = 1)
31. The equivalence conductance of 0.5 molar MgCl_2 is $980 \text{ S cm}^2/\text{equivalent}$ at 25°C . A cell with electrodes of 10 cm^2 surface area and 1 cm apart is filled with the same solution. If EMF of the cell is 25 volt and current strength is 'i' amp. Then what is the value of 'i'?
32. 2 gm of protective colloid 'X' is found to be sufficient to prevent the coagulation of 20 ml standard gold sol, before the addition of 2 ml 20% NaCl solution. What is the gold number of 'X'?
33. What is the sum of lone pairs, sigma bonds and π – bonds in one XeOF_4 molecule?
34. Two transparent glass flasks 'A' and 'B' of volume 13.2 litre and 26.4 litre respectively, connected by a tube of negligible volume. Fitted with a stopcock. Flask 'A' contains 32 gm $\text{CH}_4(\text{g})$ while flask 'B' is filled with 71 gm of $\text{Cl}_2(\text{g})$ both at 336 K. The assembly is irradiated by sunlight, when stopcock was opened. Assuming that $\text{CCl}_4(\text{liq.})$ and $\text{HCl}(\text{g})$ are the products and yield of the process is 50%. Find the final pressure in the container (in atm). [$R = 0.0821 \text{ atm-lit mol}^{-1} \text{ K}^{-1}$]

Mathematics

PART – III

SECTION – A

(One or More than one correct type)

This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

35. In a two dimensional plane, the Cartesian coordinates of points P, Q, R and S are given as (4, 4), (6, 6), (4, 8) and (2, 6) respectively. Let A denotes the region bounded by points P, Q, R and S, while B denotes the part of the region A for which any point L taken in region B is such that distance of L from nearest diagonal of region A is less than distance of L from any of the sides of region A. Then which of the following statements is/are TRUE?
 (A) The probability that a random selected point on Cartesian plane lies in region B if it is given that it lies in region A is $\frac{\sqrt{2}}{\sqrt{2}+1}$
 (B) The probability that a random selected point on Cartesian plane lies in region B if it is given that it lies in region A is $\frac{1}{\sqrt{2}+1}$
 (C) The number of elements in set $M = \{(x, y) : x \in I, y \in I, (x, y) \in T\}$ are 5 where T is a set formed by Cartesian coordinates of all the possible points lying in region B
 (D) The number of elements in set $M = \{(x, y) : x \in I, y \in I, (x, y) \in T\}$ is 1 where T is a set formed by Cartesian coordinates of all the possible points lying in region B
36. If P, Q, R, S are four points in a plane, then which of the following options is/are TRUE?
 (A) $(PS)(QR) \leq (PR)(QS) + (PQ)(RS)$ (B) $(PQ)(RS) \leq (PR)(QS) + (PS)(QR)$
 (C) $(PR)(QS) \geq (PQ)(RS) + (PS)(QR)$ (D) $(PS)(QR) \geq (PR)(QS) + (PQ)(RS)$
37. Let P(0, 3, 4) and Q(5, 12, 0) be two given points. A is a point on x-axis for which (PA + QA) is minimum. B is a point on y-axis for which (PB + QB) is minimum. C is a point on z-axis for which (PC + QC) is minimum, then which of the following statements is/are TRUE?
 (A) Equation of plane passing through A, B, C is $(1547)x + (325)y + (700)z = 2275$
 (B) Equation of plane passing through A, B, C is $(364)x + (325)y + (1050)z = 2275$
 (C) The volume of tetrahedron OABC (where O is origin) is $\frac{2275}{144} \text{ unit}^3$
 (D) The volume of tetrahedron OABC (where O is origin) is $\frac{2275}{408} \text{ unit}^3$

SECTION – A

(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

38. Let $\langle T_n \rangle$ consists of positive numbers and for any positive integer n, $\frac{T_n + 3}{2} = \sqrt{3S_n}$, where $S_n = \sum_{r=1}^n T_r$, then the remainder when T_{2024} is divided by 2023 is
 (A) 1001 (B) 3
 (C) 2021 (D) 7

39. Let $\langle U_n \rangle$ and $\langle V_n \rangle$ be two sequences and for any positive integer n , $U_{n+1} = U_n - V_n$ and $V_{n+1} = U_n + V_n$. If $U_{2024} = 2^{1012}$ and $V_{2024} = 2^{1013}$, then $(U_1 + 2V_1)$ is equal to
 (A) 1 (B) 3
 (C) 5 (D) 0
40. Let α, β, γ and δ be four roots of the equation $x^4 - 4x^3 + 5x^2 - 2x - 2 = 0$. If the value of $\frac{1}{(\alpha^2 - 2\alpha + 3)^2} + \frac{1}{(\beta^2 - 2\beta + 3)^2} + \frac{1}{(\gamma^2 - 2\gamma + 3)^2} + \frac{1}{(\delta^2 - 2\delta + 3)^2}$ is $\frac{a}{b}$ (where a and b are coprime numbers), then the value of $(a - b)$ is
 (A) 4 (B) 7
 (C) 1 (D) 9
41. Let $f : A \rightarrow A$ where $A = \{1, 2, 3, 4, \dots, 2024\}$, then number of functions f such that $f(f(f(f(x)))) = x \forall x \in A$ is
 (A) $\sum_{r=0}^{506} {}^{2024}C_{4r} \cdot \frac{4r!}{(4!)^r (r!)} (3)^r$ (B) $\sum_{r=0}^{506} {}^{2024}C_{4r} \cdot \frac{4r!}{(4!)^r (r!)} (3!)^r$
 (C) $\sum_{r=0}^{674} {}^{2024}C_{3r} \cdot \frac{3r!}{(3!)^r (r!)} (2)^r$ (D) $\sum_{r=0}^{674} {}^{2024}C_{3r} \cdot \frac{3r!}{(3!)^r (r!)} (2!)^r$

SECTION – A (Matching List Type)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

42. Consider the following frequency distribution:

x_i	2λ	3λ	5λ	6λ	7λ
f_i	4	2	3	2	1

Where, λ is a positive integer and the variance of above frequency distribution is 114. Match each entry in List-I to the correct entries in List-II

List – I		List - II	
(P)	The mean of above data is	(1)	10
(Q)	The median of above data is	(2)	18
(R)	The mean deviation about the mean of the above data is	(3)	24
(S)	The mean deviation about the median of the above data is	(4)	8
		(5)	4

The correct option is:

- (A) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (1); (S) $\rightarrow$ (1)
 (B) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (5); (S) $\rightarrow$ (5)
 (C) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (1); (S) $\rightarrow$ (1)
 (D) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (5)

43. Match each entry in List-I to the correct entries in List-II

List - I		List - II	
(P)	If the roots of $x^5 - ax^4 + bx^3 + cx^2 + dx - e = 0$ are real and positive ($a, b, c, d, e \in \mathbb{R}$), then minimum value of $\frac{a+b-c+d}{15\sqrt{e}}$ is	(1)	0
(Q)	Let a, b, c be positive integers satisfying $a^3 + b^3 + c^3 - 3abc = 19$, then the sum of all the possible distinct values of $ a - b $ is	(2)	2
(R)	If the remainder when $(13^{2023} + 21^{2023})$ is divided by 51 is λ , then the remainder when λ is divided by 11 is	(3)	3
(S)	If A is a square matrix such that $A \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $A^2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, then sum of square of elements in A is	(4)	1
		(5)	4

The correct option is:

- (A) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (4); (S) $\rightarrow$ (3)
 (B) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (4); (S) $\rightarrow$ (3)
 (C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)
 (D) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)

44. Match each entry in List-I to the correct entries in List-II

List - I		List - II	
(P)	If the number of ordered triplets (x, y, z) such that $x, y, z \in \{1, 2, \dots, 9\}$ with the condition that $z < \max\{x, y\}$ is λ , then the sum of digits of λ is	(1)	6
(Q)	The probability that a randomly selected positive divisor of 10^{2024} is an integer multiple of 10^{1350} , is $\frac{a}{b}$ (where a and b are coprime), then $(a + b)$ is	(2)	4
(R)	A triangle is formed by joining vertices of a 2024 sided regular polygon. If the number of such triangles which are not equilateral is nC_r (where $r < 10$), then the value of $\left[\frac{n}{100r} \right]$ is (where $[.]$ represents greatest integer function)	(3)	12
(S)	The number of function $f : A \rightarrow A$ where $A = \{1, 2, 3, \dots, 2024\}$ such that $f(1) + f(2) + \dots + f(2024)$ is even number are $a^{(3\lambda-1)} \cdot b^\lambda \cdot c^\lambda$ (a, b, c are prime number), then the value of $\frac{\lambda}{abc}$ is	(4)	8
		(5)	10

The correct option is:

- (A) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (5); (R) $\rightarrow$ (2); (S) $\rightarrow$ (1)
 (B) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (2); (S) $\rightarrow$ (1)
 (C) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (1); (S) $\rightarrow$ (2)
 (D) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (5); (R) $\rightarrow$ (1); (S) $\rightarrow$ (2)

45. Match each entry in List-I to the correct entries in List-II

List - I		List - II	
(P)	Let $A = [a_{ij}]_{3 \times 3}$ where $a_{ij} \in \{0, 1, 2, \dots, 2024\}$ for all $i, j \in \{1, 2, 3\}$, then the sum of all the possible values of $\det(A)$ is	(1)	0
(Q)	Let $A = \begin{bmatrix} 5 & -2 \\ 8 & -3 \end{bmatrix}$, then the sum of the elements lying along principal diagonal of A^{2024} is	(2)	1
(R)	Let $f(x)$ be a polynomial whose coefficients are integer and satisfies $f(1) = 1$ and $f(3) = 5$, then the smallest positive value of $f(0)$ that is possible is	(3)	2
(S)	If $\sum_{r=0}^{2024} r \binom{2024+r}{r} = \frac{4049!}{a!} \left(\frac{1}{b!} - \frac{1}{c!} \right)$, then the value of $(b + c - 2a)$ is	(4)	4
		(5)	5

The correct option is:

- (A) (P) $\rightarrow$ (1); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (3); (S) $\rightarrow$ (5)
 (B) (P) $\rightarrow$ (1); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (3); (S) $\rightarrow$ (5)
 (C) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (5); (S) $\rightarrow$ (3)
 (D) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (5); (S) $\rightarrow$ (3)

SECTION - B

(Numerical Answer Type)

This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

46. Let I be an interior point of ΔPQR such that $2\overline{IP} + 3\overline{IQ} + 5\overline{IR} = 0$ and areas of ΔPQR , ΔPIQ , ΔQIR and ΔRIP are denoted by A , A_1 , A_2 and A_3 respectively. If $\max.(A_1, A_2, A_3) = 5$, then the value of $A - \min.(A_1, A_2, A_3)$ is

47. The remainder when $\sum_{k=0}^{1012} \left(\frac{\binom{1012}{k}}{\binom{2024}{k}} \sum_{r=k}^{2024} \left(\binom{r}{k} \binom{2024}{r} \right) \right)$ is divided by 49 is
- (Where $\binom{n}{r}$ denotes nC_r)

48. If A and B are two square matrices of same order such that $B^3A = AB$ and $(BA)^{2024} = B^\lambda A^{2024}$ where $\lambda = \frac{a^b - 1}{c}$ (a is prime number), then the value of $\frac{b-c}{a}$ is

49. If $D_1, D_2, D_3, \dots, D_{900}$ are 900 doors and $P_1, P_2, P_3, \dots, P_{300}$ are 300 persons. Initially all doors are opened. Changing the status of all doors means closing the door if it is open or opening it if it is closed. P_1 changes the status of all doors. Then P_2 changes the status of $D_2, D_4, \dots, D_{900}$ (doors having number which are multiples of 2). then P_3 changes the status of $D_3, D_6, D_9, \dots, D_{900}$ (doors having number which are multiples of 3) and this process is continued till P_{300} changes the status of D_{900} , then the number of doors which are finally closed after this process are
50. If the value of $1^2 + 3^2 + 5^2 - 7^2 - 9^2 - 11^2 + 13^2 + 15^2 + 17^2 - 19^2 - 21^2 - 23^2 + \dots - (2027)^2$ is $-a^3 \cdot b^4$ (where a and b are coprime numbers), then the value of $(a + b)$ is
51. If area of triangle whose vertices are $(\alpha^2, \alpha), (\beta^2, \beta)$ and (γ^2, γ) is 3, then the absolute value of
- $$\begin{vmatrix} 4\alpha^2 & (\alpha + \beta)^2 & (\alpha + \gamma)^2 \\ (\alpha + \beta)^2 & 4\beta^2 & (\beta + \gamma)^2 \\ (\alpha + \gamma)^2 & (\beta + \gamma)^2 & 4\gamma^2 \end{vmatrix}$$
- is